

Curriculum Vitae - Ayush Agrawal

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Education

Present Sep 2025	Northeastern University (Supervisor: Prof. Lawson Wong) Ph.D. Computer Science Coursework: <i>Information Theory, Reinforcement Learning, Learning and Motivation (Psych), Neural Mechanics of Concepts</i>	Boston, U.S.A.
May 2022 Aug 2018	Birla Institute of Technology and Science (BITS) Pilani B.E. Electronics & Instrumentation	Pilani, India

Publications

J=Journal, C=Conference

- [J.1] **Physical Reasoning and Object Planning for Household Embodied Agents**  
[Ayush Agrawal](#), Raghav Prabhakar, Anirudh Goyal, Dianbo Liu
Transactions on Machine Learning Research [TMLR 2024]
- [J.2] **SignLoc: Robust Localization using Navigation Signs and Public Maps**   
Nicky Zimmerman, Joel Loo, [Ayush Agrawal](#), David Hsu
Robotics and Automation Letters [RAL + ICRA 2026]
- [C.1] **Sequence Agnostic Multi-Object Navigation**  
Nandiraju Gireesh*, [Ayush Agrawal](#)*, Ahana Datta*, Snehasis Banerjee, Mohan Sridharan, Brojeshwar Bhowmick, Madhava Krishna (* = Equal Contribution)
IEEE International Conference On Robotics And Automation [ICRA 2023]
- [C.2] **CLIPGraphs: Multimodal Graph Networks to Infer Object-Room Affinities**   
[Ayush Agrawal](#)*, Raghav Arora*, Ahana Datta, Snehasis Banerjee, Brojeshwar Bhowmick, Krishna Murthy Jatavallabhula, Mohan Sridharan, Madhava Krishna (* = Equal Contribution)
IEEE International Conference On Robot And Human Interactive Communication [RO-MAN 2023]
- [C.3] **Sign Language: Towards Sign Understanding for Robot Autonomy**   
[Ayush Agrawal](#)*, Joel Loo*, Nicky Zimmerman*, David Hsu (* = Equal Contribution)
IEEE International Conference On Robotics And Automation (under review) [ICRA 2026]

Research Experience

Present Sep 2025	Northeastern University Graduate Research Assistant / Advisor: Dr. Lawson Wong Working on sample efficient learning in commonsense based physical tool-use environments like virtual-tools	Boston, MA
July 2025 Nov 2024	National University Of Singapore (NUS) Research Intern / Advisor: Dr. David Hsu Worked on improving scene understanding, robot localization and open-world navigation by leveraging "navigational signs".	Singapore
June 2023	Research Intern / Advisors: Dr. Dianbo Liu , Dr. Anirudh Goyal Developed a framework and 4 commonsense QA datasets (150K QA pairs) to analyse and improve physical commonsense reasoning in embodied agents.	
June 2023 May 2022	International Institute Of Information Technology (IIIT) Robotics Research Center Research Assistant / Advisors: Dr. K. Madhava Krishna , Dr. Mohan Sridharan , Dr. Krishna Murthy Developed computational methods inspired by human cognition to enhance the performance of embodied agents in multi-object navigation, and household tidying-up tasks.	Hyderabad
May 2022 Nov 2021	University of New South Wales (UNSW) Bio-Engineering Lab Research Intern (Bachelor Thesis) / Advisor: Dr. Sridhar Ravi Designed a neural network to mimic honey bee vision based obstacle avoidance in drones.	Remote / Canberra, Australia
Aug 2021 June 2021	Indian Institute Of Technology (IIT) ARMS Lab Summer Intern / Advisor: Dr. Arpita Sinha Developed a Decentralized Multi-Drone Terrain Exploration algorithm using PX4 drones on ROS and Gazebo.	Remote/ Mumbai, India

Select Research Projects

Few-Shot Learning of Physical Tool Use

Sep'25 - Present

Advisors: [Dr. Lawson Wong](#)

- Worked on implementing information theoretic sampling techniques to map out the reward landscape using a gaussian process for VirtualTools environment.
- Explored the shortcomings of VLMs on the task of few-shot learning of physical tool use. (unable to leverage prior tries to formulate a coherent causal model of the environment)
- Currently approaching generalized tool use as a world model induction task using graph neural networks and gaussian processes.

Human inspired Navigation using navigational signs

Nov'24 - July'25

Advisors: [Dr. David Hsu](#)

- Built a large-scale Singapore navigational sign dataset with human behavioral annotations, systematically exposing failure modes of state-of-the-art VLMs.
- Proposed a principled sign-parsing framework that significantly improves over existing methods [[Preprint](#)], and extended it into a deployable topological localization system on Spot [[RAL + ICRA 2026](#)].

Physical Common Sense Reasoning

June'23 - Feb'24

Advisors: [Dr. Dianbo Liu](#), [Dr. Anirudh Goyal](#)

- Formulated a 3-step architecture to delineate physical commonsense reasoning for household tasks' object selection.
- Created human preference datasets and 4 QA datasets to analyze physical commonsense reasoning capabilities of LLMs.
- Finetuned LLMs on our datasets and analyzed the performance gain.[[TMLR'24](#)]

Embodied Multi-Object Navigation

May'22 - June'23

Advisors: [Dr. K Madhava Krishna](#), [Dr. Mohan Sridharan](#), [Dr. Krishna Murthy](#)

- Developed commonsense-oriented heuristics to optimize the search and retrieval task of multiple objects by framing the problem as a Contextual TSP.
- For static objects, developed a modular framework with an RL policy based on semantic inputs to output effective long-term goals thus enabling the robot to locate the list of objects in an optimized sequence agnostic manner[[ICRA'23](#)][[Blog](#)]
- For dynamic objects, developed a Graph Neural Network by processing Human Preference Dataset and CLIP Features to give better human commonsense aligned Object-Room Affinities and latent embeddings.[[RO-MAN'23](#)][[Intuition](#)]

Bio-Inspired Robotics

Sept'21 - May'22

Advisors: [Dr. Sridhar Ravi](#), [Dr. Puneet Mishra](#), [Dr. Sujan Yenuganti](#)

- Developed a Deep Neural Network(LSTM+CNN) to model the relationship between Geometric Optic Flow and Honey Bee trajectories. Our developed obstacle avoidance algorithm achieved an accuracy of **75%** and was verified in various simulated multi-obstacle tunnels. [[Thesis](#)][[Presentation](#)]
- Proposed and Procured institute scholarship for developing Autonomous Source Localization Drone on a DJI Tello Drone utilizing the biologically inspired Run-Tumble Algorithm[[Report](#)]

Robotics & Electronics

Jan'21 - Jan'22

Advisors: [Dr. Arpita Sinha](#), [Dr. Meetha Shenoy](#), [Dr. Puneet Mishra](#),

- Implemented a Decentralized Multi-Drone Terrain Exploration algorithm on ROS and Gazebo using PX4 drones, ensuring complete exploration in a limited number of steps [[Code](#)][[Video](#)]
- Constructed a specialized dataset featuring common UAV noises and implemented a Deep Neural Network(GAN) for effective denoising of UAV-captured images [[Report](#)]
- Designed ESP32-AWS architecture for soil condition sensing by integrating microcontrollers and sensors to enable regular data transmission to AWS Database[[Presentation](#)]

Patents

Method And System For Multi-Object Tracking And Navigation Without Pre-Sequencing, 2023 | Patent Pending

Talks

“Bio-Mimicry”

‣ A Honey Bee’s Attempt at Obstacle Avoidance [🌐]

December 2021 (BITS Pilani)

Honours and Awards

ICRA Travel Support, 2026 [🌐] Travel support to attend ICRA 2026 in Vienna, Austria

Fatima Fellowship, 2024 [🌐] Awarded compute and research mentorship to pursue ML research. (Declined)

OpenAI Researcher Access Program, 2023 [🌐] For my proposal on ”Physical Commonsense Reasoning abilities in LLMs”

BITS Pilani Undergraduate Project Scholarship, 2021 [🌐] For my proposal on ”Autonomous Odor Localization Drone”

Bronze Medal, Univeristy Physics Competition 2021 [🌐] For presenting our solution as a white paper in 48 hours

Academic Service

Teaching Assistant Reinforcement Learning Spring 2026
Reviewer IROS 2023,2024 | TMLR | ICRA 2026

Skills

Languages	Python, C++, MATLAB
Frameworks	Pytorch, Tensorflow
Tools	Git, Visual Studio
Simulators	Habitat, AI2Thor, Virtual Home, Gibson, Gazebo
	Neural Networks & Fuzzy Logic, Object Oriented Programming, Digital Image Processing
MOOC	RL by David Silver, Deep Learning Specialization by Andrew Ng